

Mini Project 1: Student Performance Analysis via PCA

Objective

To understand how **subspaces**, **span**, **orthogonal projection**, **eigenvalues**, and **eigenvectors** form the mathematical foundation of **dimensionality reduction** and **data de-noising**, applied here to student exam scores.

Background

Each student's scores across 10 exams can be represented as a vector in $\mathbb{R}^{10}$. With 12 students, we collect these into a dataset $\mathbf{X} \in \mathbb{R}^{12 \times 10}$.

StudentID	Exam_1	Exam_2	Exam_3	Exam_4	Exam_5	Exam_6	Exam_7	Exam_8	Exam_9	Exam_10
S1	80.0	73.6	81.5	90.2	72.7	72.7	90.8	82.7	70.3	80.4
S2	70.4	70.3	77.4	55.9	57.8	69.4	64.9	78.1	65.9	60.9
S3	89.7	72.7	75.7	60.8	69.6	76.1	63.5	78.8	69.0	72.1
S4	69.0	93.5	74.9	64.4	83.2	62.8	77.1	55.4	61.7	77.0
S5	82.4	76.7	73.8	72.0	60.2	67.8	70.4	85.6	78.4	57.4
S6	78.2	71.1	68.2	81.1	85.3	84.3	66.6	71.9	78.3	84.8
S7	70.2	73.1	63.9	63.0	83.1	88.6	74.3	85.0	78.6	68.5
S8	78.6	90.4	74.6	90.6	48.8	83.2	75.9	72.0	75.9	55.1
S9	72.8	78.6	89.8	69.8	66.9	70.0	84.2	78.3	69.7	80.1
S10	76.0	84.7	68.0	71.7	71.1	60.4	78.0	77.6	75.1	72.7
S11	60.8	70.8	71.6	67.0	73.4	79.0	93.9	76.7	77.6	74.3
S12	55.8	74.7	75.6	99.6	73.1	78.0	74.7	63.3	86.4	82.5

- The scores live in a high-dimensional space (10D).
- However, much of the meaningful variation (e.g., overall ability, specific strengths) lies in a lower-dimensional subspace.
- PCA helps us find that subspace by computing eigenvectors of the covariance matrix.

By projecting a student's performance vector onto the span of the top k eigenvectors, we can:

1. **Summarize** student performance (dimensionality reduction).
2. **Filter out noise** in the scores (de-noising).
3. **Visualize** performance in 2D or 3D.

Part 1: Theoretical Questions (15 pts)

Q1. Subspace & Span (3 pts)

We have 12 student vectors $\mathbf{x}_1, \dots, \mathbf{x}_{12} \in \mathbb{R}^{10}$. Explain why

$$S = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_{12}\}$$

is a subspace of $\mathbb{R}^{10}$. What is the **maximum possible dimension** of S ?

Q2. Covariance & Eigen-Decomposition (5 pts)

We form the covariance matrix

$$C = \frac{1}{n} X^T X$$

where rows of X are centered student score vectors.

1. What does an eigenvalue λ_i of C represent about variation in student scores?
2. Why are the eigenvectors orthogonal, and what does that mean about the subspace they form?

Q3. Orthogonal Projection (4 pts)

Let V_k be the first k eigenvectors. Projection of a student's centered score vector $\mathbf{x}$ is

$$\hat{\mathbf{x}} = V_k V_k^T \mathbf{x}.$$

1. Show why this is an orthogonal projection.
2. Explain why we must subtract and later add back the mean score vector in PCA-based reconstruction.

Q4. Eigenvalues & Approximation (3 pts)

Suppose the top eigenvalues are $[15.2, 9.3, 3.5, 1.0, 0.8, \dots]$.

- Compute the **total variance**.
- What is the minimum k such that the top- k eigenvalues capture at least **90% of the variance**?

Part 2: Coding Practical (10 pts)

Dataset: Use the provided `small_student_scores.csv`.

Task: Apply PCA to student scores, visualize, and reconstruct.

1. **Load and center** the student data.
2. **Perform PCA** with `scikit-learn`, keeping the top 5 components.
3. **Plot:**
 - The mean score profile.
 - The first 3 eigenvectors (as "performance patterns").
4. Pick one student at random:
 - Project their scores into the 5D PCA space.
 - Reconstruct their scores from these 5 components.
 - Compare original vs reconstructed.
5. Repeat reconstruction with only 2 components. Comment on the difference.

Coding Questions (to answer in notebook):

1. What do the top eigenvectors seem to represent? (e.g., general ability vs specific exam strengths)
2. How does reconstruction quality change from 2D $\rightarrow$ 5D?
3. Why does PCA give a compressed representation of each student?

Part 3: Extensions & AI Connections (10 pts)

1. **Denoising (4 pts):**
 - Add random Gaussian noise to a student's scores.
 - Reconstruct using only the top 3 eigenvectors.
 - Compare noisy vs denoised — why does PCA remove noise?
2. **AI Analogy (3 pts):**
 - PCA here acts like a **linear encoder-decoder**. Compare this with an **autoencoder neural network**. What is similar? What is different?
3. **Bias (3 pts):**
 - Suppose your dataset only had students strong in math-heavy exams. What might PCA's top components emphasize?
 - How could this bias affect how you interpret weaker students?

Part 4: Research (10 pts)

Examine how SVD is helpful in PCA.